

Market failure

Market failure occurs when freely functioning markets fail to deliver an efficient allocation of resources. The result is a loss of economic and social welfare. Market failure exists when the competitive outcome of markets is not efficient from the point of view of society as a whole. This is usually because the benefits that the free market confers on individuals or businesses carrying out a particular activity diverge from the benefits to society as a whole.

PRIVATE COSTS: Private costs are the costs which are incurred by the producers when they produce goods and services e.g. cost of labor, raw materials, and machines and so on. Consumers also incur private costs when they buy different goods and services. The greater the production or consumption of goods and services, the higher will be the private costs to producers and consumers. These are also called internal costs in the sense that they are borne by those who produces or consumes goods and services.

PRIVATE BENEFITS: Private benefits are the benefits which are received by the producers when they sell goods and services which they produce. The revenue which the producer receives is the private benefit. Consumers also receive the benefits when they consume the goods and services. The utility or satisfaction which they receive is the private benefit. These benefits are also called internal benefits in the sense that they are received only by those who produce or consume goods and services.

EXTERNALITIES: Externalities are third party effects arising from the production and consumption of goods and services for which no appropriate compensation is paid. Externalities cause market failure if the price mechanism does not take account of the social cost and benefits of production and consumption. Externalities occur outside of the market i.e. they affect economic agents not directly involved in the production or consumption of a particular goods or service. Since externalities do not go through the market, they do not have prices attached to them.

NEGATIVE EXTERNALITIES: Negative externalities occur when production or consumption impose external costs on third parties outside of the market for which no appropriate compensation is paid. For example consider the situations in

There is no such thing as luck, believe in hard work

Market failure

which a business dumps the waste products from its production process in to a nearby river or in which an individual litters a park or beach. Obviously, a cost is involved in these actions. When the firms pollute the water, people downstream suffer consequences. They may not want to swim in or drink polluted water. They may also be unable to catch fish as before because of pollution. In the case of littering, the people who come along after our litter has cluttered the park or the beach are the ones who bear the costs. The costs of these actions is borne by people other than those who commit these actions, the creator of the costs is not the sole bearer. The costs are not internalized by the individual or firm; they are external.

Some examples are:

1. Smokers ignore the unintended but harmful impact of passive smoking on nonsmokers or tribute to the growing costs of treating smoking-related diseases.
2. Acid rain from power stations
3. Air pollution from the road use and traffic congestion the social costs of drugs and alcohol abuse
4. The environmental damage cause by the intensive use of fertilizers in agriculture

POSITIVE EXTERNALITIES: They exist when third parties benefit from the spillover effects of reproduction and consumption. Examples are:

1. Social returns from increase investment in education and training. If one person gets educated, it benefits the entire society. The person can work as a trained worker, or he/she can be a trained professional worker at any job.
2. Positive benefits from improvements in the quality of health care and medical research.
3. Improved health outcomes arising from vaccination program. If one gets vaccinated, it prevents others form contracting the disease from that person, thus a third party is not involved in the transactions.
4. Provision of flood protection system and installation of more effective fire safety equipment

There is no such thing as luck, believe in hard work

Market failure

5. The social benefits to be derived from restored historic buildings and monuments.
6. External benefits from people usage of public libraries and museums.

SOCIAL COSTS: It is the total cost which a society bears as a result of production and consumption activity in society. It is the sum of private costs (costs to the producers and consumers of goods and services) and external costs (costs which the third parties bear as a result of production and consumption of others).

SC = PC + EC. If social cost exceeds private cost that means there will be external cost/negative externalities, for e.g. the social cost of developing tourism on a remote island is greater than the private cost because of the harm to the environment caused by the tourist.

SOCIAL BENEFITS: These are the total benefits which a society receives as a result of production and consumption activities in the society. It is the sum of private benefits (benefits received by the producers and consumers) and external benefits (benefits received by the third parties). **SB = PB + EB.** Net social benefit refers to social benefit minus social cost. Net external benefit refers to external benefit minus external cost.

CAUSES OF MARKET FAILURE AND GOVERNMENT INTERVENTION: There are several important circumstances under which markets fail to allocate resources with reasonable efficiency; some of them are:

1. **PUBLIC GOODS:** Goods that are neither excludable nor rivalrous are called public goods or sometimes called collective consumption goods. The classic case is national defense. An army of a given size protects all the nation's citizens equally no matter how many citizens are there and whether or not a particular individual pays taxes to support it. Other examples include lighthouses, street lights (it is a public good because no one can be prevented from enjoying the benefits), radio broadcasts etc. In case of public goods the benefits it confers on consumers can be extended to others at zero cost.
Because public goods are non-excludable, private firms will not provide them. The main reason for private sector not to produce public goods is that the

There is no such thing as luck, believe in hard work

Market failure

price charged must be zero. Consumption of public goods is non-rival, meaning that the consumption of a public good or service by one individual does not preclude consumption by another individual; it does not diminish the quantity for other individual. Secondly, consumption is non-excludable. This means that consumption by one individual makes it impossible to exclude any other individual from having the opportunity to consume. Of course, in this situation, private sector businesses are unlikely to consider providing pure public goods because there is no mechanism to charge the price as many consumers can take a free ride on such goods because of non-excludability. The provision of pure public goods is therefore a cause of market failure. Since market economy does not provide public goods, a government rectifies the market failure by providing the entire output. The production of public goods has to finance by the government as people are able to consume public goods without paying for them.

Another characteristic of public good is non rejectable, an individual cannot reject public good. For example, one day an individual decides that he/she is not requires the services of defense, this action does not means that defense services will now protect the whole country expect that individual.

Public goods and the free rider problem: Consumers have an incentive to not reveal their willingness and ability to pay for public goods if they believe that they will be expected or required to contribute to financing the public good accordingly by the government. After all, if the public good is supplied, it will be available to them just as it would be to anyone else because public goods are non-excludable. This is the essence of the free-rider problem: the incentive which consumers have to avoid contributing to financing public goods in proportion to their valuation of such good.

Free goods have zero opportunity cost since consumption is not limited by scarcity. Free goods have no prices and, in principle, no factors of production are required to produce them. It is not easy to think of examples. In some economies, wild fruit and berries may be gathered or some non-domesticated animals hunted for their meat. The air we breathe could also be seen as a free good. Other examples include rainfall and water in a local river.

There is no such thing as luck, believe in hard work

Market failure

Quasi-public goods: Flood control systems and lighthouses are often referred to as pure public goods. This is because they are clearly non-excludable and non-rival. Some goods that may appear to be public goods do not meet both of these characteristics in full. For this reason, such goods are known as quasi-public goods. A toll road is a typical example. It is non-rivalrous like a public good but excludable like a private good since anyone not paying the toll is excluded from using it.

Another example of a quasi-public good might be a sandy seaside beach. The beach is available to all those who wish to use it. It appears non-excludable. However, it is possible to think of ways of excluding consumers from the beach. Privately owned beaches do this. Equally, the beach is non-rival up to a point. If you are the first person on a pleasant beach on a warm, sunny day, you may continue to enjoy the beach as a few more people arrive to enjoy the benefits themselves. However, as the beach becomes crowded, space limited and other people's conversations and music become louder, you may no longer enjoy being on the beach. The beach has something of the characteristic of non-rivalry, but not the full characteristic. It is a quasi-public good.

2. **LACK OF INFORMATION:** Consumers and producers require complete information if they are to make efficient choices. In the market economy we assume that all agents in the market have perfect information about the availability of goods and services and also the prices charged by suppliers. Consumers can make purchasing decisions on the basis of full and free information on the products that they are buying. In reality all of experience information deficits which can lead to misallocation of resources. Information failure occurs when people have inaccurate, incomplete, uncertain or misunderstood data to make potentially wrong choices. Consumers might under or overestimate the private benefit from consuming a particular good or service. The classic case of this is the demand for health or education - where consumers may underestimate the long term benefit from investing in extra education or buying a specific form of health treatment.

The effects of asymmetric information: Asymmetric information occurs when somebody knows more than somebody else in the market. Such asymmetric

There is no such thing as luck, believe in hard work

Market failure

information can make it difficult for two people to do business together. Examples of asymmetric information include the following:

- A government selling mobile phone or broadcasting licenses does not know what buyers are prepared to pay for them (an auction is usually the preferred solution to this)
- A mortgage lender does not know how likely a borrower is to repay their loan in future years
- A used car seller knows about the quality of the car being sold than do buyers.

Asymmetric information can distort people's incentives to buy and sell goods and services at the right prices and as a result can lead to inefficiencies and market failure. Often market failure results from consumers suffering from a lack of information about the costs and benefits of the products available in the market place.

Government action can have a role in improving information to help consumers and producers value the true cost and benefit of a good or service. Examples might include:

- Compulsory labeling on cigarette packages with health warnings to reduce smoking
- Improved nutritional information on foods to counter the risks of growing obesity
- Anti-speeding television advertising to reduce road accidents and advertising campaigns to raise awareness of the risks of drink driving
- Advertising health screening programs / information campaigns on the dangers of addiction.

These programs are really designed to change the perceived costs and benefits of consumption for the consumer. They don't have any direct effect on market prices, but they seek to influence demand and therefore output and consumption in the long run. Of course it is difficult to identify accurately the effects of any government information campaign, be it the campaign to raise awareness on the

There is no such thing as luck, believe in hard work

Market failure

Aids issue or to encourage people to give up smoking. Increasingly adverts are becoming harder hitting in a bid to have an effect on consumers.

3. **EXTERNALITIES:** In the market economy private producers are concerned with the private costs and private benefits only i.e. they aim to maximize profits. They do not take external costs and benefits into account. When production involves external costs, the output is more than socially optimum level i.e. there is over allocation of resources. On the other hand when the production involves external benefits, the output produced is less than socially optimum level i.e. there is under allocation of resources. Both these situations lead to market failure.

Demerit goods: These are the goods which are harmful for the society and have immense external costs. Examples include cigarettes, alcohol and drugs. These are over produced and undercharged by the market economy because people are unaware of the true cost of consuming them. They are overproduced which results in an increase in supply therefore a fall in the price level, a fall in price level therefore increases quantity demanded by individuals, so there is too much production and consumption of these goods.

Cigarettes cause unpleasant smoke which is dangerous to people in the area who are forced to become passive smokers. They inflate the national health bill because both the smokers and the passive smokers get sick. But smokers will not stop, perhaps are unable to do so because they are addicted, they have the lack of information or knowledge regarding the effects of these goods.

Too many demerit goods are demanded, so the **government steps in and taxes** highly in order to reduce the consumption and to raise the revenue which is needed anyway to spend treating the sick ones. The government advertises heavily to try to persuade people stop smoking and the young not to start and is seriously considering banning smoking in all public work places. The government imposes indirect tax on the producer causing the supply curve to shift to the left. The vertical distance between the two supply curves is the per unit tax, the price of the good rises and the output falls to the socially optimum level. The burden of indirect tax on producers and consumers will depend on the elasticity of demand and supply.

There is no such thing as luck, believe in hard work

Market failure

These goods have inelastic demand, so taxing these goods won't be effective as consumers will be ready to pay higher price, only benefit will be to the government in this situation as they will receive tax revenue. It is important to note that measuring per unit tax is also difficult for the government i.e. how much to tax these goods, for that they have to do research which can increase their administrative cost and every decision of the government will involve opportunity cost. If the per unit tax is set too high, this might bring inflationary pressure and on the other hand, if too low then this will be ineffective.

Standardization: When demand for a good is price inelastic, putting a pollution tax will simply be passed on to consumers in the form of higher prices with very little reduction in output. Society is having more or less the same amount of pollution. For example, a tax on automobiles will simply increase price of automobiles without any significant cut in the number of cars and the amount of pollution remains more or less same. As an effective policy measure the government may impose standardization on automobiles as how much fuel and pollution a car engine can give out. If it exceeds the emission criteria, such a car will not be allowed on roads. These changes in the rules and regulation force the automobile manufacturers to re-design cars. The result is more efficient engines, more efficient exhaust controls, aero dynamic body, EFI etc.

Selling the right to pollute: Nature recycles a large amount of pollution via evaporation, degradation, ultra-violet rays and so on. Problem arises when society pollutes more than nature's ability to recycle and in such a case pollution becomes cumulative. In addition to this, some items stay in the environment for so many years e.g. plastic bags. The authorities of a given area can estimate the amount of pollution such an area can safely handle. This amount is then sold to local business at some given price. Any producer who does not buy the right to pollute cannot produce in that area.

Merit goods: There is a class of goods whose consumption benefits are higher for the society as compared to individuals, these goods are called merit goods e.g. education and health. When a person gets educated, the acquired knowledge and skills increase the income of individual. To society actual benefits are higher productivity, new and better products, civilized people etc. in such a case a market

There is no such thing as luck, believe in hard work

Market failure

economy under allocates scarce resources to the provision of merit goods. These goods, due to lack of information/knowledge are under consumed i.e. less demanded, which results in fall in price and provides signal to producers to under allocate the resources i.e. too little consumption and production. When talking about merit goods a government believes that consumers derive greater benefit from a good than consumers themselves realize. E.g. Education is considered to be merit good as it benefits both the recipient and society as a whole

Examples include:

- Labor training in firms; one firm may do little, as it knows that when a trained worker leaves, someone else benefits - but the first firm paid for all the training!
- Education generally
- Governments often subsidize the cost of entrance to museums because social benefits exceed private benefit.
- The growing of flowers in a private garden results in positive externalities as social benefits exceed private benefits.
- Health generally, especially in poor third world countries.
- The provision of playing field at or near schools so that the health and sporting skills of the children improves.
- Free museums and art gallery that can encourage the poor and uneducated to widen their horizons educate themselves and generally improve.

Government provides subsidies to the activities that lead to positive externalities. This reduces the costs of production for suppliers and encourages a higher output. For example the government may subsidize state health care; public transport or investment in new technology for schools and colleges to help spread knowledge and understanding. There is also a case for subsidies to encourage higher levels of training as a means to raise labour productivity and improve our international competitiveness.

Assigning the property rights: Once cause of market failure is the limited nature of property rights. A factory throws its liquid waste into the river and its smoke into the atmosphere because no one owns them. Both the river and the air are

There is no such thing as luck, believe in hard work

Market failure

common property resources. If they were in private hand, the owners would have self interest in preserving them. The contribution of each economic agent is minute, but summed over all agents, these actions degrade the resources and may cause severe long term damage.

So, the solution of the problem of market failure is to extend property rights. Property rights define who owns the property, to what uses it can be put, the rights other people have over it and how it may be transferred. By extending these rights individuals may be able to prevent other people from imposing costs on them. Ultimately property rights could be extended so completely that everyone would have the right to prevent people imposing any costs on them whatsoever or to charge them for the privilege. For example if I was disturbed by noisy Lorries outside my home, I would have the right to prohibit them from coming down my street or to charge them.

Possible problems with government intervention

There are number of reasons why government actions takes to offset market failure may result in markets working less efficiently. These include the following:

- **A lack of information:** To setting tax rates, subsidies, maximum and minimum prices a government doesn't possess accurate information about present and future market conditions. New classical economists believe the market is more efficient at discovering and transmitting information about the most efficient ways of allocating and using information about the kinds of goods and services dominated by consumer's and about the cheapest methods of production. They believe that as the economy is constantly changing governments cannot obtain the necessary relevant information at the right time.
- **A lack of continuity in government policy:** If government policy changes too frequently it will make it difficult for firms to plan ahead. For instance, if the level of subsidies is changes frequently firms may not have the confidence to proceed with investment projects.
- **Bureaucracy:** It may take time for a government to decide on a policy while implementing it may involve cumbersome procedures.

There is no such thing as luck, believe in hard work

Market failure

- **A reduction in incentives:** Government intervention distorts incentives which new classical economists argue are essential for the efficient working of markets as they provide the motivation for entrepreneurs, managers and workers to increase their effort. For example, some forms of government intervention e.g. subsidies may permit inefficient firms to survive. Here, a government may help a new firm to start up but there is a danger that the firm becomes reliant on government assistance and does not seek to improve its performance.
- **Self-interest:** On occasion politicians and civil servants may pursue their own career interest rather than society's interest. For example, a Minister who is hoping to gain employment as a director of a tobacco company when s/he retires may not support an increase in excise duty on tobacco. A government may also be pressured by special interest groups to allocate resources to satisfy their own specific interests.

ZARAK MUSHTAQ

There is no such thing as luck, believe in hard work